UPS



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 17 labs

Covers these lab items: Backup Power Supply (UPS); Online UPS 5 KVA; Power Protection; Uninterruptible Power Supply (UPS); UPS

What it is used for

A UPS provides temporary backup power and helps protect connected equipment during power interruptions. Correct installation, load management, and battery care support safe and reliable operation.

Main parts

- Input connection receives power from the mains supply.
- Output sockets or terminals supply backup power to connected equipment.
- Battery stores energy for use during power failure.
- Inverter converts battery DC power into usable AC power.
- Control panel displays operating status, alarms, and battery condition.
- Circuit breaker or protective device helps interrupt unsafe electrical conditions.
- Ventilation openings or cooling fans remove heat from internal components.

Before you start

1. Read the manufacturer's instructions and confirm the UPS rating suits the connected load.
2. Allow only trained and authorised personnel to install or service electrical connections.
3. Inspect the enclosure, power cord, plugs, terminals, and ventilation openings for damage.
4. Place the UPS on a stable, dry, well-ventilated surface away from heat sources.
5. Check that the battery is installed correctly and there are no warning alarms.
6. Confirm the connected equipment's total power demand stays within the UPS rating.

How to use

1. Switch off the UPS and connected equipment before making permitted external connections.
2. Connect the input supply to a properly rated and earthed outlet as instructed.
3. Connect compatible equipment to the designated UPS output sockets or terminals.
4. Switch on the mains supply and then start the UPS according to its instructions.
5. Check the display for normal operation, battery status, and active alarms.
6. Switch on connected equipment and verify that the UPS supports the load correctly.
7. During a power failure, save work and shut down equipment when backup time becomes limited.
8. Restore mains power and check that the UPS returns to normal charging operation.
9. Switch off and disconnect the UPS before cleaning its external surfaces.
10. Report unusual alarms, overheating, swelling batteries, or unexpected shutdowns to authorised personnel.

⚠ Safety

- Never open the UPS enclosure or touch internal terminals; dangerous voltage may remain present.
- Do not overload the UPS or connect equipment that exceeds its specified rating.
- Keep the UPS away from water, moisture, flammable materials, and blocked ventilation.
- Never short-circuit battery terminals or connect batteries with incorrect polarity.
- Do not remove or replace internal batteries unless trained and authorised to do so.
- Treat swollen, leaking, unusually hot, or damaged batteries as hazards and report them immediately.

Care & storage

- Keep ventilation openings clear and wipe the exterior with a dry cloth after disconnecting power.
- Inspect external cables and connections regularly for damage, looseness, or overheating.
- Follow manufacturer instructions for battery testing, replacement, and preventive maintenance.
- Arrange authorised servicing if backup time decreases significantly or alarms persist.

Common mistakes to avoid

- Connecting equipment whose total power demand exceeds the UPS rating.
- Blocking cooling vents or placing the UPS in a damp area.
- Opening the enclosure or handling internal batteries without authorisation.
- Ignoring warning alarms, unusual heat, or reduced backup time.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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